Swiftrail Logistics Database Schema

Swiftrail Logistics Database Overview

This database is designed to support the logistics operations of Swiftrail Logistics Company. It stores information about employees, customers, shipments, drivers, vehicles, warehouses, and shipment status history. The design follows a relational database model with primary and foreign keys to maintain data integrity.

Roles

The Roles table defines the different job roles within Swiftrail Logistics Company. Each role represents a specific level of responsibility and access permissions, ensuring that employees can only perform actions appropriate to their position.

Attribute Name	Data Type	Constraints
role_id	INT	PK
role_name	NVARCHAR(50)	UNIQUE, NOT NULL
description	NVARCHAR(255)	NULL

Employees

The Employees table stores information about company employees, including their personal details, department, and assigned role. It is used to identify users and manage role-based access within the system.

Attribute Name	Data Type	Constraints
employee_id	INT	PK
full_name	NVARCHAR(100)	NOT NULL
email	NVARCHAR(100)	UNIQUE
password_hash	NVARCHAR(255)	NOT NULL
department	NVARCHAR(100)	NULL
role_id	INT	FK → Roles

Customers

The Customers table stores customer information, including contact details and addresses. Each customer can create one or more shipment requests.

Attribute Name	Data Type	Constraints
customer_id	INT	PK
full_name	NVARCHAR(100)	NOT NULL
phone	NVARCHAR(20)	NOT NULL
email	NVARCHAR(100)	NULL
address	NVARCHAR(255)	NOT NULL

Warehouses

The Warehouses table stores information about the company's warehouses where shipments are received, processed, and dispatched before delivery.

Attribute Name	Data Type	Constraints
warehouse_id	INT	PK
warehouse_name	NVARCHAR(100)	NOT NULL
city	NVARCHAR(100)	NOT NULL
address	NVARCHAR(255)	NOT NULL

Vehicles

The Vehicles table stores details of all delivery vehicles, including their license plate, model, carrying capacity, and operational status.

Attribute Name	Data Type	Constraints
vehicle_id	INT	PK
plate_number	NVARCHAR(20)	UNIQUE
model	NVARCHAR(50)	NULL
capacity	DECIMAL(8,2)	NULL
status	NVARCHAR(30)	CHECK

Drivers

The Drivers table stores information about delivery drivers and the vehicles assigned to them. It also tracks each driver's availability for shipment assignments.

Attribute Name	Data Type	Constraints
driver_id	INT	PK
full_name	NVARCHAR(100)	NOT NULL
phone	NVARCHAR(20)	NULL
vehicle_id	INT	FK → Vehicles
status	NVARCHAR(30)	CHECK

Shipments

The Shipments table is the core table of the database. It stores shipment information, including the customer, assigned driver, warehouse, delivery addresses, shipment status, and delivery dates.

Attribute Name	Data Type	Constraints
shipment_id	INT	PK
customer_id	INT	FK → Customers
driver_id	INT	FK → Drivers
warehouse_id	INT	FK → Warehouses
pickup_address	NVARCHAR(255)	NOT NULL
delivery_address	NVARCHAR(255)	NOT NULL
status	NVARCHAR(30)	CHECK
created_at	DATETIME	DEFAULT GETDATE()
expected_delivery	DATETIME	NULL
delivered_at	DATETIME	NULL

Shipment_Status_History

The Shipment_Status_History table records every status update made to a shipment. It provides a complete history of shipment progress, including who updated the status and when the update occurred.

Attribute Name	Data Type	Constraints
history_id	INT	PK
shipment_id	INT	FK → Shipments
status	NVARCHAR(30)	NOT NULL
updated_by	INT	FK → Employees
updated_at	DATETIME	DEFAULT GETDATE()

Relationships

Parent Table	Child Table	Relationship
Roles	Employees	One-to-Many
Customers	Shipments	One-to-Many
Drivers	Shipments	One-to-Many
Warehouses	Shipments	One-to-Many
Vehicles	Drivers	One-to-One
Shipments	Shipment_Status_History	One-to-Many
Employees	Shipment_Status_History	One-to-Many
Parent Table	Child Table	Relationship
Roles	Employees	One-to-Many
Customers	Shipments	One-to-Many